



The Impact of Zinc Oxide Nanoparticle and Carnosic Acid on Quality Beef Meat under Refrigerator Storage

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ABSTRACT

Background: This study plane to evaluate the effectiveness of zinc oxide nanoparticles and carnosic acid in prevent beef meat from oxidative deterioration and spoilage bacteria under refrigerated storage to improve meat quality and increase shelf life.

Methods: Samples were divided into four groups: T1 (control, without any additives), T2 treated with (1% zinc oxide nanoparticles), T3 treated with (2% zinc oxide nanoparticles) and T4 treated with (2% carnosic acid). All beef samples were stored under refrigeration at 5°C and analyzed at different storage interval period (0, 3, 4, 7 and 12 days).

Result: Our result indicated that Peroxide value, thiobarbituric acid (TBARS), myoglobin content and total bacterial count significantly much reductions ($P<0.05$) in all treated samples compared to the control during the storage period. The peroxide values and total bacterial count were high in the control groups after 12 days of storage, indicating lower preservation compared to the treated groups.

Key words: Beef meat, Carnosic acid, Zinc oxide nanoparticle.

INTRODUCTION

Meat is an important source for human nutrition and contains on high nutrition value such as; proteins, fatty acids, minerals and vitamins (Alnori, *et al.*, 2022 and Devatkal *et al.*, 2025). It is affected by many factors that cause of flavour change such as chilled, microbial growth and lipid oxidation that lead to generation of sulfur compounds on the surface of the meat and secondary lipid oxidation products (Needham *et al.*, 2019). Also, lipid oxidation could have negative impacts on meat quality; to protect them and increase shelf-life, many food additives are often used. Addition, today's consumers are increasingly replacing artificial food additives with natural alternatives (Carrapiso *et al.*, 2023 and Kanaan *et al.*, 2025).

Nanoparticles that has the ability to offer a wide range of uses and improved food chain processing to improve food safety, quality control and increase the shelf life of foods (El Asuoty *et al.*, 2023). It's one of the promising and useful antibacterial agents that could possibly be applied in therapeutics (Kalra *et al.*, 2022). Also, using food as food additives is one of the most importance uses of nanotechnology in food and meat to preserve colors and prevent spoiling (Lamri *et al.*, 2021) and Biswas *et al.* (2022).

Zinc oxide nanoparticles is stable antibacterial agent that inhibitor growth bacterial cells through many mechanisms involving different chemical species, such as reactive oxygen species and Zn^{2+} (Hakeem *et al.*, 2020 and Arulnathan *et al.*, 2023). However, it can play an significant role in decrease the risk of food contamination with pathogens and increase shelf life of foods (Saafan *et al.*, 2019). ZnO is generally recognized as safe by the U.S. Food and Drug Administration (El Asuoty *et al.*, 2023).

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Rosemary plant is one of the plant extracts that have an antioxidant and antimicrobial properties (Wali *et al.*, 2025). Also, have phenolic diterpenes that are natural functional ingredients such as carnosic acid, carnosol, rosmanol, rosmariquinone, rosmaridiphenol, caffeic acid and ursolic acid. About 90% of the antioxidant properties of rosemary extract were found to be from their carnosic acid and carnosol constituents. Its antioxidant activity is about seven times higher than that of synthetic antioxidants (Peñaranda *et al.*, 2021). The aim of study systematically evaluating and comparing the effects of zinc oxide

nanoparticles and carnosic acid on the microbiological quality, physicochemical properties and shelf-life stability of beef meat during refrigerated storage.

MATERIALS AND METHODS

Sample collection

This study conducted from October 2025 to December 2025. Fresh beef samples were purchased from retail meat processing in Babylon-Iraq. Immediately all samples were kept in ice-box and stored at 5°C. All samples were transferred to the Public Health Laboratory at the College of Veterinary Medicine, Al-Qasim Green University, carnosic acid and zinc oxide nanoparticle were obtained from Sigma Aldrich Chemical Company (USA).

Sample preparation

Fresh beef samples were chopped into small pieces using a plate with 4mm holes. The samples were divided into 4 equal groups. Control group (T1) (without any added), while the other 3 group (T2 and T3) were mixing with 1%, 2% of zinc oxide nanoparticle respectively while (T4) was add 2% of carnosic acid. After mixed, fresh beef samples of 50 gram were placed in polyethylene film and store in refrigerated at 5°C for 12 days.

Peroxide value

The peroxide level of samples was determined according to (Ahmad, *et al.*, 2023). Approximately 5 gram of beef samples were mixed with anhydrous sodium sulfate (0.1N) and chloroform (30 mL) then centrifuged at 10,000 rpm at 4°C and filtration by filter paper and add two mL of saturated potassium iodide solution and 30 mL on supernatant and left for two min, through intermittent shaking. However, add 100mL of distilled water and two mL of freshly prepared starch solution (1%) and titrated sodium thiosulfate (0.1N) until became colorless of a non-aqueous.

Determination of TBARS value

Estimated of TBARS level in fresh beef sample according to (Saeed and Abdulwahid, 2023) with few changes. Add 3 gram of samples to 25 mL of PCA (3.86%) and mixed for 20 second during an ultra turrax homogenizer (UTH) and then take 2 ml from filtrates after filtering of homogenates and incorporated into 2 mL of TBA (20 mM) in distilled water. These solutions were then kept for 16 h at room temperature to obtain absorbance. Absorbance at 531 nm was determined using UV-Vis spectrophotometer and TBARS level was expressed as mg of malondialdehyde (MDA)/kg of sample.

Myoglobin value

The myoglobin concentration was measures according to (Faris *et al.*, 2024) weight 10 gram of each samples and homogenized with distilled water (90 ml). Then homogenized after take 10 g of this mixture and added distilled water (15 ml). Then filtered of this mixture with filter paper No. 1

and measure the sample using of spectrophotometer wavelength of 525.

Microbial counts

For microbial evaluation, total bacterial count was performed according to the (Ahmad *et al.*, 2023). A ten gram quantity of beef sample was aseptically extracted from the stored stock sample. Each of the samples were ground using blinder and then weighting ten grams of the sample and transferred into a sterile container containing 90 ml of 1% peptone water at room temperature then serial dilution 10^{-1} to 10^{-6} of the samples and take 1 ml of dilution plated on nutrient agar at 37°C for 24 hours.

Statically analysis

The data were analysis the mean±standard deviation (Mean±SD) of the experiments data. Two-way analysis was using to statically estimated of all data by (ANOVA) using SAS for analysis of variance significant P value ($P<0.01$) (SAS.SAS/STAT, 2010).

RESULTS AND DISCUSSION

Peroxide value

Fig 1 presents the effect of different concentrations of zinc oxide nanoparticles (ZnO-NPs) and carnosic acid on the peroxide value of beef meat during refrigerated storage. A significant decrease ($P<0.05$) in peroxide values was observed across the storage periods (0, 3, 4, 7 and 12 days) in both treated and untreated samples. Significant differences ($P<0.05$) were also detected among the various treatment doses of ZnO-NPs and carnosic acid throughout the storage period.

The control group (T1) recorded the highest peroxide values, measured at 2.614 ± 0.17 , 3.70 ± 0.15 , 4.55 ± 0.23 , 6.05 ± 0.19 and 8.90 ± 0.19 meq/kg on days 0, 3, 4, 7 and 12, respectively. In contrast, treatment with 2% carnosic acid and 2% ZnO-NPs (T4) resulted in markedly lower peroxide values of 0.58 ± 0.05 , 1.89 ± 0.06 , 2.92 ± 0.01 , 3.62 ± 0.16 and 5.74 ± 0.20 meq/kg over the same storage intervals. Similarly, samples treated with 2% ZnO-NPs alone (T3) showed reduced values of 0.85 ± 0.08 , 1.25 ± 0.05 , 2.04 ± 0.01 , 3.91 ± 0.31 and 4.72 ± 0.17 meq/kg.

Our study demonstrated a significant ($P<0.05$) reduction in peroxide values in beef samples treated with 1% and 2% ZnO nanoparticles (ZnONPs) and 2% carnosic acid compared with untreated control during refrigerated storage at 4°C. This confirms the strong antioxidant potential of both ZnONPs and carnosic acid in delaying lipid oxidation. These results are consistent with Naveena *et al.* (2013), who found that carnosic acid at different concentrations (22.5 ppm and 130 ppm) effectively suppressed lipid oxidation in fresh and cooked minced buffalo and poultry meat.

ZnO-NPs are recognized as safe and efficient additives for increase the shelf life of packaged foods. According to the United States Food and Drug Administration (US-FDA),

ZnONPs are classified as generally recognized as safe (GRAS) Priyadarshi *et al.* (2017). In addition, prior studies have shown that chitosan films incorporated with 2% ZnO nanoparticles exhibit superior functional properties compared with pure chitosan films. These composite films are biodegradable, eco-friendly and possess strong antimicrobial potential, making them suitable for food-packaging applications Priyadarshi *et al.* (2017).

Lipid peroxidation commonly increases during meat handling, transportation and storage, leading to rancidity and deterioration of sensory quality, particularly in beef due to its relatively high fat content Alizadeh-Sani *et al.* (2020). In the current study, the initial peroxide level of fresh beef was 0 meq/kg; depending on the type of packaging or treatment used through storage the peroxide level which gradually increased. Similar trends were reported by Priyadarshi *et al.* (2021), noting that packaging materials significantly influence oxidative stability during storage.

It is significant to note that the acceptable limit for peroxide value in fats and oils should not exceed 10 meq/kg fat, as recommended by Al-Majidi *et al.* (2015) and the Iraqi Standard Specifications Central Organization for Standardization and Quality Control (1987). The treated

samples in this study remained well below this threshold throughout the storage period, further confirming the effectiveness of ZnONPs and carnosic acid in maintaining oxidative quality and extending the shelf life of beef meat.

Thiobarbituric acid value

The findings presented in (Fig 2) indicate a significant decrease ($P < 0.05$) in thiobarbituric acid (TBA) values between the different treatments of zinc oxide nanoparticles (ZnO-NPs) and carnosic acid compared with control group across all refrigerated storage periods. However, no significant differences ($P > 0.05$) were found among the various treatment concentrations of ZnO-NPs and carnosic acid within each storage time (0, 3, 4, 7 and 12 days).

The control group (T1) exhibited the highest TBA values, recording 0.54 ± 0.02 , 0.67 ± 0.02 , 1.00 ± 0.06 , 1.38 ± 0.06 and 2.19 ± 0.08 mg MDA/kg on days 0, 3, 4, 7 and 12, respectively. In contrast, the lowest TBA values were observed in samples treated with 2% carnosic acid (T4), which showed 0.38 ± 0.02 , 0.51 ± 0.02 , 0.71 ± 0.08 , 0.83 ± 0.03 and 1.16 ± 0.06 mg MDA/kg over the same storage intervals.

Similarly, samples treated with 1% ZnO-NPs (T2) recorded TBA means of 0.44 ± 0.01 , 0.55 ± 0.03 , 0.79 ± 0.05 ,

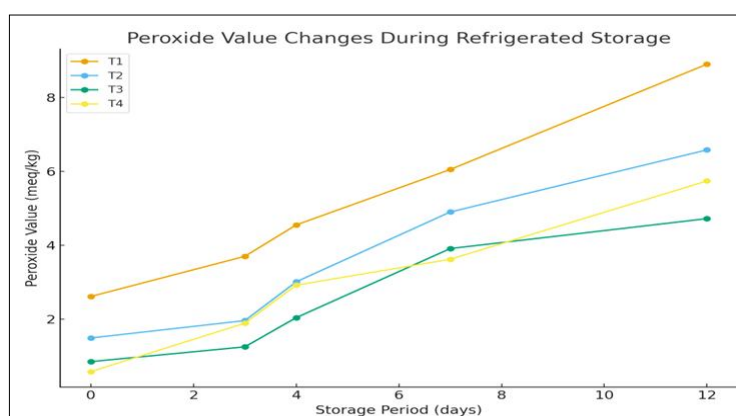


Fig 1: Effect of zinc oxide nanoparticles and carnosic acid on peroxide value in beef meat at refrigerator storage.

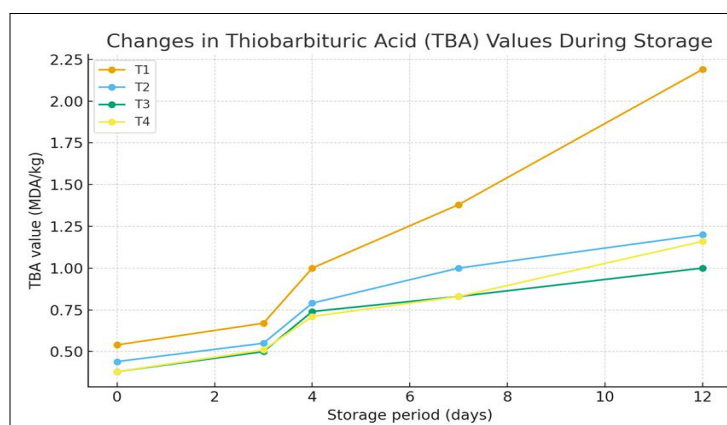


Fig 2: Effects of zinc oxide nanoparticle and carnosic acid on thiobarbituric acid value in beef meat at refrigerator storage.

0.83±0.08 and 1.20±0.06 mg MDA/kg, while those treated with ZnO-NPs in treatment T3 showed corresponding values of 0.38±0.02, 0.50±0.04, 0.74±0.06, 0.83±0.08 and 1.00±0.07 mg MDA/kg during refrigerated storage.

The thiobarbituric acid reactive substances (TBARS) values decreased in samples treated with zinc oxide nanoparticles (ZnONPs) and carnosic acid compared with the control during refrigerated storage, indicating an effective reduction in lipid oxidation. These findings are consistent with Moran *et al.* (2012), who reported lower TBARS values in lamb meat when animals were fed low doses of carnosic acid. Similarly, Zhang *et al.* (2010) demonstrated that carnosic acid effectively suppressed malondialdehyde (MDA) formation in sunflower oil, while Redondo-Cuevas *et al.* (2019) showed that the addition of carnosic acid at various concentrations to rapeseed oil significantly reduced MDA levels compared with untreated samples. These studies support the strong antioxidant capacity of carnosic acid in inhibiting secondary lipid oxidation products.

The effect of ZnONPs on TBARS values has been discussed with some variability in the literature. Ramacharyulu *et al.* (2014) reported that the antioxidant activity of ZnONPs may decline because their potential to generate reactive oxygen species (ROS). However, several studies have shown beneficial antioxidant effects of ZnONPs in meat systems. Newman *et al.* (2009) observed a significant reduction in lipid oxidation in meat samples treated with ZnONPs compared with the control. Saeed and Abdulwahid (2023) also reported minimal increases in TBARS values in ZnONP-treated samples during storage (0-7 days). This effect was attributed to the antimicrobial properties of ZnONPs, which reduce bacterial load on the meat surface, thereby slowing bacterial-induced lipolysis and decreasing the formation of oxidation products.

The reduction in TBARS levels may also be linked to the ability of ZnONPs to interfere with microbial enzymes responsible for the decomposition of unsaturated fatty acids, thus limiting the creation of both primary and secondary lipid oxidation products. Hatab *et al.* (2023)

further demonstrated that ZnONPs decreased MDA levels in broiler serum, reinforcing their role in reducing oxidative stress. Likewise, Hassan *et al.* (2023) reported enhanced antioxidant capacity in chickens supplemented with plant-derived ZnONPs at 70 ppm.

Overall, the present results indicate that both carnosic acid and ZnONPs exhibit strong antioxidant and antimicrobial properties, contributing to lower TBARS values and improved oxidative stability of meat during refrigerated storage.

Total bacterial counts

The findings presented in (Fig 3) demonstrate the effect of zinc oxide nanoparticles and carnosic acid, applied at different concentrations, on the total bacterial count of beef meat during refrigerated storage. The untreated samples (T1; control) exhibited a significant ($p<0.05$) and progressive increase in total bacterial count throughout the storage period, with mean values of 6.22±0.20, 7.01±0.24, 7.42±0.16, 7.61±0.20 and 8.28±0.26 log cfu/g at 0, 3, 4, 7 and 12 days, respectively.

In contrast, beef samples treated with zinc oxide nanoparticles and carnosic acid showed a significant ($p<0.05$) lower in bacterial growth at all tested concentrations compared with control group. The lowest bacterial counts were observed in samples treated with 2% carnosic acid (T4), which recorded mean values of 5.63±0.04, 5.56±0.03, 5.60±0.33, 5.36±0.21 and 5.42±0.20 log cfu/g across the same storage intervals. Similarly, treatment with 2% ZnO-NPs (T3) effectively suppressed microbial growth, with mean bacterial counts of 5.53±0.06, 5.52±0.03, 5.34±0.18, 5.27±0.21 and 5.28±0.25 log cfu/g during the storage periods.

The total bacterial count showed a marked decrease in all meat samples treated with ZnO nanoparticles (ZnONPs) and carnosic acid during refrigerated storage at 4°C. This finding aligns with Al-Alwani, (2017), who reported that carnosic acid effectively reduced the total bacterial load in minced meat. The antimicrobial action of carnosic acid is supported by Ojeda-Sana *et al.* (2013), who explained

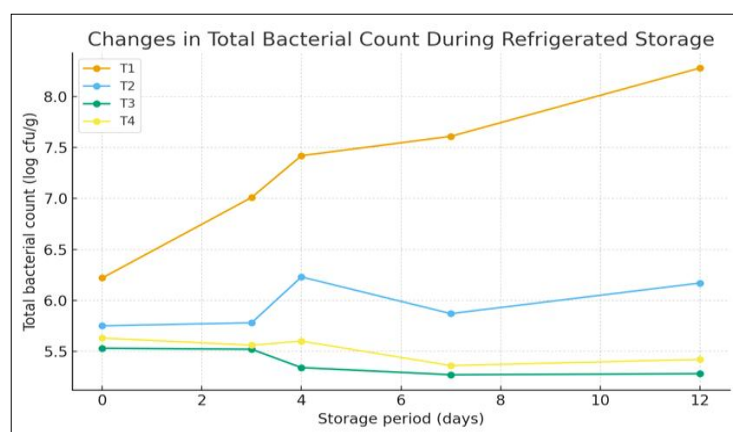


Fig 3: Effects of zinc oxide nanoparticle and carnosic acid on total bacterial count in beef meat during refrigerator storage.

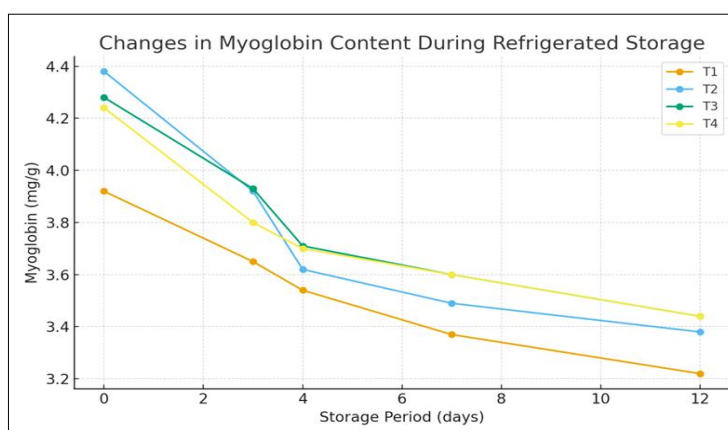


Fig 4: Effects of zinc oxide nanoparticle and carnosic acid on myoglobin value in beef meat during refrigerator storage.

that it disrupts the permeability of bacterial cell membranes, causing structural deformation, functional loss and ultimately cell death.

The current results also agree with Alqahtani (2025), who observed a significant reduction in total bacterial count in broiler chicken meat treated with ZnONPs. Several studies have highlighted the strong antimicrobial potential of ZnONPs. Kadhim *et al.* (2024) demonstrated that ZnONPs exhibit broad-spectrum antibacterial activity against foodborne pathogens responsible for food poisoning. Similarly, Mohammadi *et al.* (2019) reported that poultry breast meat coated with ZnONPs showed a significant reduction ($P < 0.05$) in bacterial load compared to untreated samples.

Collectively, these findings indicate that both ZnONPs and carnosic acid contribute to improved microbial quality and extended shelf life of meat by inhibiting bacterial growth through their antimicrobial mechanisms.

Myoglobin value

The results show in (Fig 4) there was a significant ($p < 0.05$) decrease in myoglobin content in both untreated and treated beef meat samples during refrigerated storage. The untreated control group (T1) exhibited a gradual decline in myoglobin values, with mean concentrations of 3.92 ± 0.05 , 3.65 ± 0.02 , 3.54 ± 0.02 , 3.37 ± 0.02 and 3.22 ± 0.01 mg/g at 0, 3, 4, 7 and 12 days of storage, respectively.

Samples treated with 1% ZnO-NPs (T2) showed higher initial myoglobin retention but similarly exhibited a gradual decrease over time, recording values of 4.38 ± 0.04 , 3.92 ± 0.07 , 3.62 ± 0.02 , 3.49 ± 0.01 and 3.38 ± 0.01 mg/g during the storage period. Likewise, treatment with 2% ZnO-NPs (T3) resulted in mean myoglobin concentrations of 4.28 ± 0.04 , 3.93 ± 0.05 , 3.71 ± 0.04 , 3.60 ± 0.02 and 3.44 ± 0.02 mg/g across the corresponding time points.

In comparison, beef samples treated with 2% carnosic acid (T4) maintained myoglobin values of 4.24 ± 0.04 , 3.80 ± 0.06 , 3.70 ± 0.06 , 3.60 ± 0.06 and 3.44 ± 0.02 mg/g throughout the 12-day storage period.

Moran *et al.* (2012) reported that supplementing fattening male Merino lambs with carnosic acid for five

weeks resulted in improved meat color and a noticeable reduction in color oxidation compared with the control group. Similarly, Naveena *et al.* (2013) demonstrated that the application of carnosic acid at concentrations of 22.5 ppm and 130 ppm in minced buffalo meat as well as in fresh and cooked chicken meat significantly influenced pigment stability, particularly myoglobin and metmyoglobin. Overall, both zinc oxide nanoparticles and carnosic acid helped preserve higher myoglobin levels compared with the control group, indicating an enhanced protective effect against oxidative pigment degradation during cold storage.

CONCLUSION

The findings of this study clearly demonstrate that both zinc oxide nanoparticles (ZnONPs) and carnosic acid are highly effective in enhancing the oxidative stability and microbiological quality of beef during refrigerated storage. Treatments containing 1% and 2% ZnONPs as well as 2% carnosic acid important decrease peroxide values, TBARS formation and total bacterial counts compared with the control, indicating strong antioxidant and antimicrobial activity. Carnosic acid effectively inhibited lipid oxidation and stabilized meat pigments, while ZnONPs reduced microbial load and slowed the growth of spoilage bacteria.

Overall, the combined results confirm that ZnONPs and carnosic acid can be applied as promising natural and nano-enhanced preservation agents to increase shelf life, maintain quality and improve the safety of fresh meat during cold storage. Their effectiveness and compatibility with food systems suggest valuable potential for use in modern meat preservation and packaging technologies.

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Conflict of Interest

There was no conflicts of interest.

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